PROJECT REPORT

TASK 33: SIMPLE LINEAR REGRESSION USING PYTHON

1. Abstract

This project focuses on building a Simple Linear Regression model using Python, Pandas, NumPy, Matplotlib, and Scikit-learn.

A Salary vs Experience Dataset containing 20 fictional employee records is used to predict annual salary based on years of work experience.

The project demonstrates dataset preparation, input and target selection, training and testing data separation, model training, prediction, and evaluation.

A regression line is also created to visualize the relationship between years of experience and annual salary.

2. Introduction

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions.

Linear Regression is a supervised machine-learning algorithm used to predict numerical values.

Simple Linear Regression uses one independent variable to predict one dependent variable.

In this project, years of experience is used as the independent variable, while annual salary is the dependent variable.

The objective is to understand how a simple regression model learns the relationship between these two variables and predicts salary values.

3. Objectives

The main objectives are:

- To understand Simple Linear Regression.
- To prepare a numerical dataset.
- To select one input feature and one target variable.
- To divide data into training and testing sets.

- To build and train a Linear Regression model.
- To generate predictions on test data.
- To evaluate the model using regression metrics.
- To visualize the regression line.
- To understand the regression coefficient and intercept.

4. Tools and Technologies

Python: Used as the primary programming language.

Pandas: Used for dataset creation, loading, and manipulation.

NumPy: Used for numerical operations.

Scikit-learn: Used to create, train, and evaluate the Linear Regression model.

Matplotlib: Used to visualize the data points and regression line.

Google Colab: Used as the Jupyter Notebook development environment.

CSV: Used to store the salary dataset and prediction results.

5. Dataset Description

The project uses a manually created Salary vs Experience Dataset containing 20 fictional employee records.

The dataset contains two columns:

Column	Description
years_experience	Total years of employee experience
annual_salary_inr	Annual salary in Indian Rupees

The dataset is designed to demonstrate a positive and approximately linear relationship between experience and salary.

The values are fictional and do not represent actual company salaries.

6. Understanding Simple Linear Regression

Simple Linear Regression is a supervised machine-learning algorithm that models the relationship between one independent variable and one dependent variable.

It uses a straight-line equation:

Y = mX + C

Where:

- Y represents the predicted output.
- X represents the input feature.
- m represents the slope or regression coefficient.
- C represents the intercept.

The algorithm attempts to find the best-fitting straight line that minimizes the squared prediction errors on the training data.

7. Feature and Target Selection

The project uses the following variables:

Independent Variable (X): Years of Experience.

Dependent Variable (y): Annual Salary.

Years of experience is selected because it represents an employee's professional experience.

Annual salary is selected because it is the numerical value the model is intended to predict.

Only one numerical input feature is used, making this a Simple Linear Regression problem.

8. Data Preparation

The dataset is created using Python and converted into a Pandas DataFrame.

The CSV file is saved and loaded using Pandas `to_csv()` and `read_csv()` functions.

The dataset is inspected to understand its dimensions, column names, data types, and numerical characteristics.

The input feature and target variable are separated before training.

9. Training and Testing Data Separation

The dataset is divided using Scikit-learn's `train_test_split()` function.

An 80:20 split is used.

Dataset	Percentage	Records
Training set	80%	16
Testing set	20%	4

Total	100%	20

The training dataset is used to teach the model.

The testing dataset is used to evaluate predictions on observations not used for training.

10. Model Development

The Linear Regression algorithm is imported from:

`sklearn.linear_model`

A model object is created using `LinearRegression()`.

The model is trained using the training dataset through the `fit()` method.

The trained model learns the relationship between years of experience and annual salary.

11. Model Parameters

After training, the regression coefficient and intercept are extracted.

The approximate results are:

Parameter	Value
Regression coefficient	40,040.72
Intercept	2,65,184.74

The regression equation is:

Annual Salary = 265184.74 + (40040.72 × Years of Experience)

The coefficient indicates that the model estimates an increase of approximately ₹40,040.72 in annual salary for every additional year of experience.

The intercept represents the estimated salary at zero years of experience according to this fitted equation.

12. Salary Prediction

The trained model is used to predict salaries for the testing dataset through the `predict()` method.

The predicted values are compared with actual salary values.

Experience	Actual Salary	Predicted Salary
1 year	₹3,00,000	₹3,05,225.46
2 years	₹3,40,000	₹3,45,266.17
16 years	₹9,00,000	₹9,05,836.22
18 years	₹9,80,000	₹9,85,917.66

The predictions demonstrate how the trained regression model estimates numerical salary values.

13. Model Evaluation

The model is evaluated using four regression metrics.

13.1 Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual and predicted values.

Result: ₹5,561.38

13.2 Mean Squared Error (MSE)

MSE calculates the average squared prediction error.

Result: 31,029,545.81

13.3 Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and measures prediction error in the original salary unit.

Result: ₹5,570.42

13.4 R-squared (R^2)

R^2 measures how much of the variation in the target variable is explained by the regression model.

Result: 0.9997

The model shows a very high R^2 value on the test observations. This is expected because the educational dataset was constructed with an almost linear pattern.

14. Regression Line Visualization

A scatter plot is created to display actual salary observations against years of experience.

The fitted regression line is added using the predictions generated by the trained model.

The visualization includes:

- Years of experience on the X-axis.
- Annual salary on the Y-axis.
- Training observations.
- Testing observations.
- Fitted regression line.

The upward direction of the regression line indicates a positive linear relationship in the sample dataset.

Insert the actual salary_regression_line.png image here.

15. New Salary Prediction

The trained model is also used to estimate the annual salary of a new employee with 12 years of experience.

The estimated annual salary is:

₹7,45,673.35

This demonstrates how a trained regression model can be used to generate predictions for new input observations.

16. Key Observations

The project provides the following observations:

1. The dataset contains 20 fictional employee records.
2. Years of experience and annual salary have a positive linear relationship.
3. The model estimates that salary increases by approximately ₹40,040.72 for each additional year of experience.
4. The regression model produces four predictions on the testing data.
5. The model achieves an R^2 score of approximately 0.9997 on the testing dataset.
6. The regression visualization shows an upward trend.
7. The model can estimate salary for a new employee using the supplied experience value.

17. Applications

Simple Linear Regression is useful in:

- Salary prediction.
- House price estimation.
- Sales forecasting.
- Business analysis.
- Financial forecasting.
- Demand prediction.
- Trend analysis.
- Numerical prediction problems.

18. Limitations

The project uses a small, manually created dataset containing 20 fictional employee records.

The dataset has been designed to demonstrate a nearly linear relationship, so the high evaluation score should not be generalized to real-world salary prediction.

Actual salary is affected by several other factors, including qualifications, job role, location, industry, and skills.

Using experience as the only input feature may therefore be insufficient for accurate real-world salary prediction.

19. Future Enhancements

The project could be extended by:

- Using a larger real-world salary dataset.
- Including multiple relevant input features.
- Developing Multiple Linear Regression.
- Comparing different regression algorithms.
- Performing cross-validation.
- Evaluating performance on independent data.
- Developing a salary prediction application.

20. Conclusion

This project demonstrates how Simple Linear Regression can be used to predict a numerical target using one numerical input feature.

A model is developed to predict annual salary using years of work experience.

The project covers data preparation, feature selection, training and testing data separation, model training, prediction, evaluation, and regression visualization.

The exercise provides practical understanding of supervised machine learning and forms a foundation for more advanced regression techniques.

21. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. NumPy Documentation: https://numpy.org/doc/
3. Scikit-learn Documentation: https://scikit-learn.org/stable/
4. Linear Regression Documentation: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html
5. Matplotlib Documentation: https://matplotlib.org/stable/

22. Project Links

GitHub Repository: https://github.com/venky1318/Simple_Linear_Regression